

ActInf Livestream #017.0 ~ Information flow in context-dependent hierarch

Livestream | Information flow in context-dependent hierarchical Bayesian inference | 2:05:11

hello and welcome everyone

to the active inference lab and to the

active inference live stream

we're here in active inference live

stream 17.0

and it's march 3rd 2021

i'm daniel and i'm here with two of my

colleagues who can introduce themselves

blue

my name is sarah davis i'm a person

who's interested in things i'm a student

in

leibniz university in a master's program

in philosophy of science and i've

been an engineer and a bunch of other

things so i'm just generally interested

in how it's all connected

i'm blue knight i'm an independent

research consultant based out of new

mexico

cool well we're here for a fun

discussion so thanks both for joining

at the active inference lab we're an

experiment in online team communication

learning and practice related to active

inference you can find us

at our links and socials this is a

recorded and an archived

and a hastily produced live stream so

please provide us with feedback so that

we can

be improving on our work all

perspectives

and backgrounds are welcome here and

we'll follow good live stream etiquette

so here we are in paper number 17
and we've just completed our first
quarterly active inference lab
round table so check that out on our
channels if you haven't
and today we're going to be setting the
context
for 17.1 and 17.2
the paper that we're discussing in 17 is
information flow
in context dependent hierarchical
bayesian inference by chris fields and
james f
glazebrook it was published in october
2020
and what's funny is that the dot zero
videos have always been about
context set in context and we even wrote
in the previous videos
the video's introduction and its context
it's not a review or the final word
and then it got really meta with this
paper because it's a paper about
context the punchline or a punchline of
this paper
is that we can integrate various
approaches from mathematics mostly
to achieve a scale-free formulism that
leads to an interpretation
of nested systems where communicating
systems
are engaged in active inference and
distinguished by informational or
statistical patterns
or they're engaged in something that's
related to active inference let's say
because it's pretty
out there with what it is and it's a

challenging but also
a fun paper so we're gonna
start with the goals and the claims and
the abstract and
at that point it might be like wait what
with the roadmap
headers or with abstract claims it's
like driving through
a country where you don't speak the
language perhaps
and that's okay because the bulk of this
video in the keywords are going to be
going through
and kind of unpacking a lot of the key
terms that are gonna
matter for even just understanding the
lay of the land
and then we've pulled out a few other
key topics some quotations
didn't go as much into the formalisms of
the
paper itself but we're looking forward
to doing that with
the authors in 17.1 and so we're just
looking forward to that
in 17.1.2 we'll be discussing the same
paper
so read it and save and submit your
questions and get in touch if you want
to participate and also we're looking
forward to your
chats uh during this presentation giving
us a little
support along the way so here we go
with the goals and claims of the paper
under discussion so here are
um the goals and the claims so maybe
blue do you want to read the author's

goals and claims

sure so the goals in the paper are

twofold

first to model intrinsic or true

contextuality

using the general category theoretic

methods of two spaces and channel theory

and second to employ this formulation to

reconstruct hierarchical

bayesian inference in a

context-dependent way

it goes on to say here we have

formulated an approach to intrinsic

contextuality in general category

theoretic terms

a set of observations exhibits intrinsic

contextuality

if no cocoon can be constructed over the

observables that produce them

oh cone cocoon who knows

how to say it um so they're doing a

couple of things

here they have two main goals so it's

great that they're laying it out

clearly the first one is about context

and they want to be

having a more advanced way to talk about

context and it turns out that they're

going to be doing that

using category theory and two spaces and

channel theory

all of which we're going to be going

into at least a little bit and then

secondly they're going to be linking

all of this stuff that they just did up

here with context choose

channel theory category theory and

they're combined combining that with

context dependent

bayesian inference we've talked a lot
about active inference and bayesian
inference

and so this is sort of the hinge where
we connect some of these ideas
to active inference and then this is
a quote from their conclusion which is
that they have formulated
a new way of modeling contextuality
in very general category theory terms
specifically relating to this co-cone
construction

but the category theoretic term allows
them to move between a few
different areas that we're going to be
walking through

today so here we go with the abstract
i'll read the first of the three slides
of the abstract and anyone can give any
thoughts or just
start reading the second part of the
abstract

recent theories developing broad notions
of context and its effects on inference
are becoming increasingly
important in fields as diverse as
cognitive psychology

information science and quantum
information theory and computing
here we introduce a novel in general
approach to the characterization of
contextuality
using the techniques of two spaces and
channel theory viewed as general
theories of information flow
this involves introducing three
essential components into the formalism

events conditions and measurement
systems

any thoughts either of you two or
someone can just start reading the
second part of the abstract

let's go to the second part of the
abstract i think we'll have a lot of
time for um

off-the-cuff remarks soon

so sarah do you want to go for the
second part of the abstract

uh sure incorporating these factors in
relationship to conditional
probabilities leads to information flows
both in the setting of two spaces and
channel theory

the latter provides a representation of
semantic content

using local logistics from which
conditionals can be derived

we employ these features to construct
cone cacone

diagrams commutativity commutativity
of which enforces inferential coherence
um

with these built a scale-free
architecture incorporating a
bayesian-like hierarchical structure
in which there is an interpretation of
active inference in markov blankets
nice and then blue how about finish out
the abstract

we compare this architecture with other
theories of contextuality which we
briefly review

we also show that this development of
ideas conveniently accommodates negative
probabilities

leading to the notion of signed
information flow and address how quantum
contextuality
can be interpreted within this model
finally we relate contextuality to the
frame problem another way of
characterizing a fundamental limitation
on the observational and information
inferential capabilities of finite
agents
perfect so that is the abstract that's
how the authors
wanted to present and communicate their
work
to a broad audience of specialists and
non-specialists
and again we're going to be going into
basically all the words that might be
non-obvious in this abstract and the
keywords that come into play
that's what we wanted to be unpacking
today so
we're gonna look over their roadmap
because just like the abstract is how
they want to communicate
what they did the roadmap is how they've
chosen to organize what they've done
and get us from a to z and then we're
going to be
really going into this road trip
and unpacking what some of these key
terms are
it starts with an introduction which i
guess is helpful context
and then one of the focal ideas of the
paper
is this chew space or the chew
construction which we're going to be

talking about the history of
some examples some related topics and
just right off the bat when we saw that
the
examples of two spaces and chew
morphisms included
topological spaces so that's all of
network theory
and graphs and things like that so
that's one big area
but we're going to be talking about two
spaces as something that connects
all of network theory and topological
space to
types as processes probability spaces
bayesian belief networks
event space structure shoe flows kovac
libra
divergence and the free energy so
we're going to be popping up a level
from things that are already
pretty abstract but it's going to be
done in a way that's
quite useful especially for formulating
context
and it's actually really interesting
because a lot of times people will say
that these
um topics in the examples that they're
general but they actually don't have
enough context
for example you can make a social
network but that wouldn't be the total
context of the world if you only have
some of the users on the social network
so maybe there's a way where we're going
to be able to step up
from some of our previous conceptions

and
introduce context potentially by default
this leads the authors to a discussion
of
probability and how to use probabilistic
models
for contextuality for modeling context
and this comes to a recently derived
framework called contextuality by
default
which will be a topic we dive into
section 4 goes into some issues related
to
measurements and kind of the before
during and after
of measurements because you got to set
things up and calibrate and then you got
to make the measurement
at a certain moment and then you have to
analyze it in the context of a certain
system so it's like before during and
after the measurements
in section 5 they head into channel
theory
and information channels and this was
really fascinating
because it's quite distinct from shannon
information theory
in some ways that people might not
expect and
i had never heard this discussed
alongside shannon information theory
it's like
here's information theory and it's good
at abc but it's not good in these three
things
that's where the discussion ended i
didn't know that there was another area

that kind of picked up where shannon
left off in a sense
and that's related to this idea of an
infomorphism
as well as classifiers conditional
probabilities and this cocon
diagram thinking about information
channels
and flows of information leads to this
formulation of context compliant so
context
modeling or following hierarchical
multi-level
multi-scale bayesian networks and
first this section has a lot of
technical detail but it involves
making a connection between bayes
networks
approaches to probabilistic modeling and
statistics and inference
with context this leads to
a sort of conclusion or a
climax section where the hierarchical
bayesian networks that were just
introduced in six are brought into
the context of two specific uh practices
which is number one the idea of active
inference i highlighted it because
that's kind of
our lab that's what we're here to be
thinking about and then two
was the frame problem which is something
that's also
really interesting to think about it's
going to be the key motivator
so although they put it um last with
active inference in the frame problem
we're actually going to have those

topics closer to the front of the presentation
so people who are listening to this from the active inference perspective or from more generally the frame problem perspective they're going to understand what motivated us to be studying the paper the way that we're doing it and hopefully what motivated chris to select this paper when he suggested it to be discussed during his visit that's the sort of uh naked road map that's the turn left at this part and turn right at this part but there was a a narrative road map that the authors actually put out and their own words are on the right so you can see that in the paper or here but on the left i just thought of this is like the semantic road map as far as the flow of the paper and i just extracted this semantic flow from the syntax that they wrote not super knowing what all these topics are but just re-phrasing what they had put previously these authors had been researching quantum and thermodynamics of separable systems in this paper they started with a recap of two spaces and choomorphisms which suggests that it's a foundational topic for the paper they then reviewed contextuality paradigms and the diversity of approaches that model

contextuality
they then took a formal
uh diversion into
this idea into various ideas of
probability theory
above my detailed understanding but they
took an extension
in that direction they then turned to
the
semantically richer methods of channel
theory
and they're using channel theory as an
even more general approach
to inference rather than the syntactic
information theory like shannon
information theory which we'll get
to they wanted something a little bigger
thinking about
these kinds of uh local logics
of information channels leads to some
new ways
that they also combined ideas and
then they visualize this
slash represent it as this cone
cocon diagram as they say as a natural
scale
representation of bayesian inference so
scale free just means that
it doesn't have an a priori scale it's
not in inches or
in kilometers it's just at the level of
abstractness
and then they have the claim that is
proven
that the failure of the cccd the cone
cocone diagram
to diagrammatically commute is a
condition of the measurement operator's

non-co-deployability

so it's just sort of like if you have

two shapes and then they fit together

what they represent fits together and if

you have two shapes and the puzzle

pieces don't fit together

there's something mathematical that the

shapes represent isn't fitting together

in the way that it would as if these two

shapes fit together

i don't know what non-co-deployability

is

but it's like a puzzle piece if it were

co-deployable

and then the part that i think will be

of special interest to a lot of our

lab and colleagues is they're gonna

recover

this new theoretical and inferential

role for the markov blanket

it's the locus at which free energy is

minimized

part one and part two the epistemic

barrier that keeps contextual

information hidden from the observer

so when i saw that i was like whoa it

sounds fascinating because it's two

distinct and critical roles of a blanket

construction

and then i thought so contextual

information is hidden from the observer

because a lot of times we might think

about context as being like what's

around us

but it we'll see if that's actually the

way that context is being

used can i jump in for a second daniel

yep

so i think that i the way that i
understood um co-deployability
like measurement operator code
deployability like you can't
at the same time measure position
and velocity that was like one example
that they gave in the paper so like
because those two things like you can't
if you're moving you're not at a stable
position right so so it's
those those two measurements are not
co-deployable you can't measure
those two things at the same time and
that's kind of what they meant like
about
measurement code deployability i think
cool thanks for that
here's to why it matters before we even
jump into the keywords which will be the
bulk of our
journey and this is quotes from the
authors
while the results presented here are
primarily technical so
just take a breath if it's worrying you
because they wrote the paper so we don't
have to and it's a technical paper and
that's what we're
learning by doing and exploring the
paper is technical and abstract
but now we're looking for where we can
go it's like a car is technical
but it lets you get somewhere so where
are we going and why will it matter
these results open a path to application
in a number of areas where contextuality
has largely been neglected but is
important in practice

that sounds really important because it sounds like there's um systems that are unknown unknowns we don't even know what to model but there might be a large category of systems where we think we've modeled it well but then there's this problem where context keeps on entering the picture or something that we didn't model throws our local model off that we spent time and energy on so if we could find a way to respectfully interface with contextuality across different areas that would be really important well what are those areas one of the models that they're interested in just like fristen and a lot of active inference researchers is neuroscience and so they suggest that the current approach they're taking takes a previous model which is the global neuronal workspace model and that's actually related to what ryan smith and christopher white will be talking to us about in um number 18. they're going to make that model contextuality compliant and provide the building blocks needed to incorporate context switching and attention switching into global neuronal workspace models so it's kind of these two cool parts of the brain which is that it can spiral in and really hone in

with attention but then it can also
move attention so it's like side to side
and
in and out and that sort of context and
attention switching
is not dealt with well by our current
models which might be about attention or
inattention
to a stimulus but they're not actually
modeling what
the regime of attention is um
another then there's a formal result
which might matter to some
people or others that contextuality can
always be associated with
this sort of non-community kind of
related to the puzzle pieces the context
we just mentioned
and then um this is interesting from a
more probability theoretic point of view
our results further support the authors
of the contextuality by default
in showing that the distinction between
quantum and classical probabilities
lies not in any ontological difference
but rather in what has been explicitly
labeled
so this was really mind-bending because
a lot of times
it's like quantum you know theory of
small things where there's an observer
that has an effect
in the outcomes of how the system is
modeled or set up
and then um the sort of disinfo
debate is like who's an observer or does
it matter if it's a conscious deserve
you're going to go down a rabbit hole

because there aren't simple answers
there but the system of the particle
it turns out depends on how you
structure the context of the experiment
just as is the case for macroscopic
entities
so these were some questions that so so
in other words is there
a continuum of
uh tiny things and bigger things with
context as being what is distinguishing
the behavior that can seem quite
different across scales
so it's not that electrons are magical
and they can do
wave particle but nothing else exists in
a similar regime
it's not that we're going to get the
same diffraction of people walking
through a doorway
but then again maybe if it's set up the
right way there's more similarity
than not so a few opening questions
before we jump into the keywords just
how will we connect across formal and
informal or just
plain different senses of the word
context
and then what are some of these cases
where
whether they said it in the paper or not
where contextuality has been largely
neglected but it's important in practice
so even within an informal context
definition
what's in an area where potentially
context is not taken into account so
i'll start with one and then either of

you could
go maybe cultural context or
for understanding geopolitics or
individual context for
an individual relationship so it does
matter but when we approach those sort
of individual or collective level
problems as if we can ignore
context again just using it informally
we're not really operating in the space
that the system is in
we're operating in a reduced or a
distorted model
that we're projecting uh hey daniel i
i don't know if uh this is gonna mess up
your video setup but
if you're able to go into present mode
on your slides it would make the whole
thing a lot more readable for a small
screen
in the live stream which you can look at
and mute in the background
it looks different okay um
but yeah we've got a lot of small text
on these slides
yeah it's um i don't know it's so worth
unpacking
that last bullet point it's um there's
just a lot there
is is how so are you reading um
when i read this um you know that he's
saying ontological difference versus
what's explicitly labeled
i immediately just think well of course
because you're you know you're either
looking at small scale or you're looking
at macro scale but you think he's saying
something

something a little more subtle
um i would love to have a quantum
expert but just speaking with our friend
jason larkin he has often pointed to
um these sorts of almost like classical
meets quantum crossover situations
where you get the classical quote
behavior
from when you define the full system
with the quantum observer
then it behaves let's just use the twin
slit for example
the system behaves a certain way when
you define it a certain way with the
observer in the picture but when you
don't define the observer
you're just talking about a different
system
and so that might not be something that
only
electrons exhibit
it might be a feature of certainty and
uncertainty
in measurement and that would be various
kinds of measurements
it wouldn't have to be just things that
were smaller than
a molecule okay
that's how you're reading it yeah that
makes sense
but that's my non-technical um physics
for biologists
take yeah any thoughts on this blue or
we can go to the keywords
so i do have a little bit of thoughts
and so it's tied into the quantum but
it's also tied into
back in theorem 7.1 so you had skipped

over this word that i think is kind of
critical and important
um in the previous bullet point that
says our main form of result
theorem 7.1 shows that intrinsic
contextuality
can always be associated with
non-commutativity so like in that
non-commutative that means like you
can't
construct the big cocoon diagram right
like when it's not when it doesn't
commute
so i think that it's important to um
highlight that that kind of
contextuality um
is like you know where they were
describing the context like for
uh something that's beautiful like you
know a beautiful
what right like a beautiful leaf is not
the same as a beautiful person
if that leaf looked like a person it'd
be a really ugly leaf or if a person
would like leaf it'd be a really
ugly person right like so so beautiful
it's always good like that context
dependent a beautiful what like
beautiful does not always have the same
referential um meaning and so i think
that this
um is is really important and ties right
into the quantum
aspect of things right like how they
were talking about the
alice and bob um with the machine right
like so
it's always um you know when you can

measure

like it's always what is explicit

context right so the quantum is tied

into

explicit context like when we have a set

of variables

that we measure for every circumstance

at precisely the same exact time and

under precisely the same conditions

that is extrinsic context right and it's

different and separate from

intrinsic contextuality sorry i just

wanted to point that out

you might be saying something that i'm

yeah it's great thank you blue it's

um you might be saying something that

that that i'm already about to say

and um but or maybe just triggered my

thought but

yeah this intrinsic versus extrinsic the

way what it made me think of was that

um you know one thing is quantifiable

and the other thing

isn't it's like a gradient that's

essentially what you're saying yeah okay

it's great thank you fun well

let's get to the keywords this was

definitely one of our

um you know even though you look at the

keywords you think oh it's the abcs

i mean how hard is it gonna be no it was

really fun

and a good learning experience for all

of us to go through these topics

and includes active inference

and then bayesian inference which are

things that we've definitely talked

about before

as well as markov blankets a little bit
however the other words were basically
all just even new to learning about not
just in
active inference so that's um where we
kind of started this one and that's why
we want to just
include people in the conversation and
just see where everyone is at
in learning these topics so
contextuality
choose space channel theory information
flow
local logic cone cocoon diagram
frame problem and then it wasn't a key
word but it's kind of related is
concurrency and concurrent
processes and i just always want why is
the second c capitalized
let's motivate the paper um
not with what chew spaces are which is
how the paper
begins if you were to read it but let's
kind of motivate it
with the problems and the context that
we're looking to be applying in which is
the frame problem
and active inference and thinking about
something that most of us are more
familiar with which is bayesian
computation
so the frame problem is the problem of
using their words the problem of
circumscribing what does not change when
an
action is performed viewed broadly it's
the problem of circumscribing
what is relevant in a situation it's

kind of like if you want to say well
what are all the connections in the
social network
it's the shadow of what are all the
non-connections
because those two answers are arrived at
at the same time
and if you have no false positives and
no false negatives it's the same to know
what has changed versus what hasn't
changed so when we're thinking about
scenarios
those two are equivalent um
instead of just reading this for a first
pass um
what did either of you how would you
summarize the frame
problem or what is something that makes
it valuable or important
for study
okay think about it and then raise your
hand while i go through these
the issue with the frame problem and the
reason why people
are working on better solutions is that
let's say we want to understand uh what
does or doesn't change when action is
performed so we're making a self-driving
car and we want to understand how
something changes or not
when the scene changes um
in order to look for all the changes aka
what hasn't to change
it's really hard to know when to stop or
what has changed
there's just an exponentially large
number of things to check
and you can't brute force it so there

are heuristic
solutions so solutions that aren't exact
but get you on the right track
however the heuristic solutions which
often work extremely well like frame
differencing
like if something doesn't move between
two slides you just cancel it out and if
it does move you're going to do some
other algorithm on it
it's a good heuristic but it assumes
coherent context
it assumes that what is observed could
in principle be expanded
to include all that there is to be
observed so if it's a image and it's a
64 by 64 image and i do frame
differencing
and we're only studying the image then i
have captured everything that's changed
but when there are systems with
intrinsic contextuality
this coherence assumption is violated if
there is intrinsic contextuality
i.e some of the observables are
non-co-deployable
the current context cannot even in
principle be expanded to include
everything that's observable so blue
what do you
think about that or the frame problem
so one thing that um i thought was cool
about the frame problem is like
that they um so they had this corollary
8.1 i don't know if you have this up
there but when they're when they're
discussing the frame problem
it says a distributed inflammation

information flow system
can be informationally unencapsulated
only in absence of
intrinsic contextuality but so so what
that really meant to me
it was they said again they reiterated
practically that this means that proving
the absence of contextuality in a domain
requires discovering all of the
information that is relevant
to solving problems in that group in
that domain so the frame problem can
only be solved if it has already been
solved
like it's like having ground truth data
like if you're trying to predict
something like you can only predict
if you've been given a ground truth to
to predict
from or by um do you need me to repeat
that danielle you
caught it i just yes
interesting maybe just like has it been
solved like it has been solved like in
the past present or the future and we're
just like figuring it out again or
something or what does it mean
right like so it's essentially like you
can't go you can't go forward solving it
and this goes into and i know we're
gonna get into the computing
analogy later on in the in the paper the
reservoir computing but it goes into
that like
without ground truth data your neural
network is useless
like like you can't predict the future
based on something that if you don't

have the
the context for it right and not this
intrinsic contextuality but you have to
have like
like a real contextuality like know all
the variables that are
important for the for the problem um
anyway sorry that's that's my thoughts
on the frame problem
cool and at the bottom here um how would
domain specific or general solutions or
simply better heuristics lead to real
world implications
well we can probably think of a lot of
software and hardware and hybrid systems
that need to ask themselves what is
relevant in a situation
so that's going to be a recommendation
engine a self-driving car
whatever it is we want to be modeling
relevant aspects of a sim of a situation
so
that is the big and it also has
implications that each person is going
to
have their own take so you know put it
in live chat or the comment but
it's a cool topic of course there's
going to be new places of bias
that'll be kind of interesting too yep
any other thoughts on frame problems
sarah no that's it
so bayesian inference
another big topic but um we can start it
with a fun
little meme so on the top right it says
okay gang
let's see what deep learning really is

and this is like scooby dooby doo where
are you
uh meme genre and so we're gonna find
out what deep learning is at the end of
the episode
we unmask the villain oh it's rev bass
so this is sort of hinting at the idea
that bayesian inference
or bayesian thinking bayesian statistics
is underlying
what deep learning is what is
bayesian inference so there's two ways
we're going to show it here there's the
equation
in its sort of um most simple form
on the top left and then a more colorful
unpacked
version that's made a little bit closer
to natural language
there's equations and then also we can
look at it in a graphical way
and what the equation as well as the
graphical
um representation are getting at is the
idea that there's a prior
which means before and so there's a
prior expectation
and it is some type of distribution it
could be
tight or wide it could be different
types of things and there's no
uninformative prior even if it's flat
like i'm not sure if it could be a zero
percent or 100 chance or it's all equal
that's not the same thing as unbiased so
this really gives a better lens to talk
about
bias and implicit or explicit bias

because we can actually talk about
priors

being updated rather than who is or
isn't coming to the table with no
prior expectations that person doesn't
exist

priors um are transformed through
measurement into posterior
estimates and so it's like reality is
providing stimuli

and then the prior gets dragged along
either

a little bit if the learning rate is low
or a lot if it's going to be
an overwhelmingly powerful stimuli it
gets pulled a little bit or a lot
to the data and um

what else would you say about that blue
um nothing else i mean i think that
that's

that's the deal um

pretty well described what about you
sarah i mean the only thing i thought
about

it's a little vague um i mean i was just
looking at um some youtubes on koopman
embedding and model discovery
and and when i see these bays diagrams
i'm often like okay well they have some
data where they presuppose
you know what what is connected to what
prior and

with the uh with this this other method
that i was looking at it it seems like
that's not the case you know they just
feed it into this gigantic
structured ai um but so i'm just
thinking about bayesian inference in a

lot of different ways
cool and we can also ground bayesian
inference in a
history and a developmental trajectory
of statistics which is to say more
memes so first here's just one quick
meme here's bayes saying
i thought the volume of a high
dimensional hypersphere
would be evenly distributed throughout
just funny stuff
truly funny yeah exactly and on the
right side
is a smbc comic just
joking about how when data comes in your
model is updated
but that doesn't mean it's going to be
an adaptive model
so there's a lot of ways to go about
integrating new data
into priors you could have fallacious
priors you could fallaciously update you
could make the wrong
measurement you could interpret the
measurement wrong so
saying we did it bayesian is like saying
you know i built the car out of metal
it's like what else do you want to know
does it work is it safe so we can't let
the buzzwords
um confuse us we just want to follow up
with what the buzzwords are
and here in the sort of stereotypical
um human evolution meme
there's um with a little extra labeling
there's a progression
from the and theta here is the model
parameters

or just the model and then x is like the
data or the observable
so the a priori homo apriorius
just is walking around with a model
here's how the world is i
what data just it's the model this is
what's likely
homo pragmaticus is sort of again it's
not an actual evolutionary or
creationist um parallels just to me
the data is the only thing that's
considered what's the data
show me the facts facts don't lie facts
don't care about your feelings it's just
the facts it's not even about the model
and then we get to homo the frequentest
so the frequentist is asking the
statistical question
what is the likelihood of the data given
the model
so what for example um a p-value
is coming from like a t a t test
is like what is the likelihood of this
height data given the model that the two
classrooms don't have different heights
and so you can say that's 0.01 p value
so there's such and such um likelihood
for this data
under that model so we can reject that
model
and then that's all this whole point
with the rejection of
null models and H_0 versus H_1 and
rejecting things and the whole idea of
falsification at the statistical level
but also
at the philosophy of science level oh
one beautiful piece of data could ruin

the theory and you just can disprove it
this whole idea of disproving you have
the model in the background
explicitly or implicitly and then new
data comes in you go is this likely or
not
and then if it's unlikely you think that
it casts doubt on your model
homo sapiens here is thinking about the
the
likelihood or modeling both the data and
the model
but then the sort of advanced or at
least
farthest right homo bayesiansis
is thinking about the likelihood of the
model given the data
which actually is what we want to know
we don't need to know the likelihood of
all board positions and all possible
chess strategies we want to know
the likelihood of certain things
happening given what we're observing
which can often be quite limited or
confusing
so not to over analyze meme but yeah
okay
so can we wait i want to interject
really quick so we were talking earlier
about um frequentest and bayesian
statistics prior to
pushing the you know broadcast button on
the live stream and you know i was
thinking about the movie 50 first dates
right like i don't know if you guys have
seen the movie but she wakes up every
day and can't remember anything except
like before her car accident

so the model's there it's existing right
it's like she has the model
but and then just the model never
updates so every day like you know
it's you just go based on the model but
then there's no there's no update and so
like you know the bayesian
would be like if every day you updated
the model while you were sleeping at
night and i think like
you know that's a huge like like lead-in
to the bayesian brain
like framework right is that like what
we're doing while we're sleeping really
are we updating our model like is this
why we need less sleep as we get older
and we get stuck in a decade like
you know you're stuck in your favorite
decade because you're not updating your
model
as much or as well i don't remember the
movie too well but
didn't the other character have memory
across the dates so it's almost like an
alternative title it could have been you
know
when a frequentist and a bayesian date
50 times
that one definitely got left on the uh
clipping room before
so um bayesian networks are
taking this bayesian inference idea and
putting it in a graphical
or a topological framework so it's
representing
variables in the nodes there's other
graphical models but just for the ones
we're showing here

the nodes are random variables
and then conditional dependencies are
shown with
arrows or with edges so this is like a
very general one on the left it's like a
is just having a conditional dependency
with b
and it's directed in this case this one
right here
is actually a bayesian network called a
neural network
this is a single layer neural network
where one input
kind of unfolds to all these hidden
layers just one here and then it goes to
an output layer
but this is the structure of neural
networks so
everything that we're talking about like
neural networks all kinds of deep
learning are
encapsulated and then on the top right
from par at all
2018 is the example of active inference
and so here is the progression that we
walk through in our model stream
but the simplest bayesian network that
we can imagine between a hidden state
and an observable is you have s the
hidden state
and then there's some sort of p which is
the probability of an observation given
the hidden state
of oh the observable so that's like
hidden state and then you're getting
observables
and then you're trying to model the
hidden state given the observables in b

that same process happens through time a
changing hidden state
is emitting observables at different
time steps still a bayesian graphical
model

c introduces control theory with
 π that's why our lab logo is a little
pie

because it's action playing into the way
that states are modeled to evolve
through time

d takes another level of complexity and
allows actually

policy so still we see this motif with
 π influencing how s_1 , s_2 and s_3
change

but at each time point let's take this
one down into s_1

at each time point it's being evaluated
how a policy is going to play out at s_1 ,
 s_2 and s_3 . and notice that at s_1 , s_2 and
 s_3

there's an inference on s_1 , s_2 and
 s_3 and so

um i'll re-enable my camera don't worry
um

the uh the trajectory of action is what
is being estimated

so it's not just doing state estimation
at each of those time points it's
estimating a trajectory of action and
this idea of

action trajectories is going to come
back into play in a
kind of interesting way

let's just take one more second more
generally to think about active
inference and

what it is why we're here where the
markov blanket is
in active inference we have internal and
external states and it's a scale-free
framework because these internal states
could be small or large in scale
it's outside the level of specifying a
specific scale
and these internal and external states
are separated by
a blanket and as we're going to show in
just a second
this markov blanket concept is just the
total layer of insulating nodes
such that the internal and the external
states have a certain type of
independence relationship from each
other
and then the innovation we'll get to
where first thing came into play
so that's active inference just um as a
way to integrate action and perception
um with sensations influencing internal
states internal states including
generative model estimates of the world
and policy estimations
implements action and action can
influence external states which also is
undergoing its own
dynamics which we might think of as
actually contextuality
any other thoughts on active inference
okay so
here's where we go from bayes networks
to markov blankets
this was part of our discussion on
actin 14 and the idea of
making a markov blanket was introduced

by
pearl 1988 and where fristen's
innovation occurred and this is i think
from
mel andrews's paper in 14 is that
pearl's original construction of the
markov blanket
was subdivided into two kinds of nodes
sensory states nodes whose influence is
directed towards the blanketed node
under interest x and active states which
are nodes influenced by
 x so we can have an organismal metaphor
with sense and action
but a little bit more generally sense is
like incoming
of the system and action is outgoing
and the markov blanket is the total set
of all the
ones that totally give you all the
inputs and all the outputs of the system
so stating the conditional dependencies
is the same thing as stating all the
conditional independencies
just like we're showing where the social
network who's friends with who is the
same question as who's not friends with
who
who is there is the same question as who
is not there
and then to put markov pearl and fristen
on a continuum
you have markovian properties
by andre markov and other mark hopes
actually
who are working in a more mathematical
frame they don't have access to
computers

perl brings in bayesian statistics as well as computational approaches in the second half of the 1900s and then the innovations of fristen were to divide

those markov nodes the markup blanket nodes into

sense and action nodes which enables this cybernetic approach to input a generative model internal states and then

outgoing uh action states in service of maintaining a non-equilibrium steady state

and then though it's an ongoing and open and developing area with how we formalize these things

the idea is that by this kind of partition of the blanket happening we'll be able to have um modeling of non-local dependencies like latent causes which are not observed in the world

and therefore have cybernetic agents in niches that are enacting and embodied and encultured in a really rich way and then just to show one more figure this is from um

wanja's paper with carl fristen which we talked about just a few weeks ago and just showing that you have internal states that influence external uh influence active states then there's a line with external and so here's the blanket here

in the pink dashed line is the blanket states and then there's the internal states okay any thoughts on active

inference or can we go to
context so
again putting the sort of end parts of
the paper up closer to the beginning
the context in practice the first
context introduced is the idea of active
inference in the paper
and the motivation for active inference
here and we can definitely talk to the
authors more
is that observers are constantly probing
their environments through action
influenced by hidden states and
time-sensitive policies
so these are some of the challenges that
beset
context agnostic or unspecified context
modeling is when the environment is
changing
how does the action policy update does
it just re-evaluate at every single
moment
as if it was like fresh looking at a new
environment but how do you have these
sequences of action
that might involve sequential
modifications of the environment
and uncertainty in the action policy
and the big question just like you know
kindergarten
is what actions are appropriate what
actions will reveal aspects of the
environment
important for defining and reasoning
within a context that's the frame
problem what actions will reveal that a
context has changed
so we want to know which of the three

cups the ball is under
and so there's like this complex thing
that's happening as we guide our
action by changing the context to
resolve our uncertainty
about actions which we can then get more
information on
and active inference tells us that
there's two ways to minimize expected
free energy
which is that we can modify internal
state parameters of our generative model
through learning and development or
external states can be modified through
action or um their own endogenous
changes
so active inference we're kind of going
even
more abstract than some of the other
papers and just thinking about this
question
of if you're an agent who's operating
under uncertainty
modifying the niche but you don't know
whether you should modify or explore
or wait and all these things are coming
together
we want to set a frame that's big enough
to encompass that
any thoughts on that okay
let's turn to context more generally so
that was active inference in context
they write contextuality in the behavior
of complex systems has traditionally
been regarded
as a practical problem of limited
experimental control that could in
principle

be eliminated by obtaining more uniform
experimental subjects and achieving
better control
of experimental conditions this
traditional view of contextuality as
merely a practical limitation is
howeverly
increasingly under strain this kind of
resonated with me as an experimentalist
because
in biology you'd get the sense like well
if it was a clonal animal and we just
had the
light dark cycle the same and the same
animal handler picked it up and there
was no pipetting air
everything was the same you'd get the
same result
and it was really exciting to think
about
um to what extent is that like a
faith-based position
and actually tiny amplifications in the
system you have two clonal animals but
if there's like a bifurcation point
then unless you truly have every single
part of the system
you're always going to be subject to
these intrinsic contextualities
so that's kind of cool to think about
and then here in the conclusion
context is often used to designate what
is neglected when an observation is made
or an action undertaken
environment is used in a similar way so
that's how maybe informally people think
about context like right now one of my
contacts is like

this cup of water it's in my environment
it's in my niche it's in my local area
like their context is used
like that like what is around something
um and the formalism of a theory like
context by default
challenges us to acknowledge that
context is always there whether attended
to or not
so we're thinking about context in sort
of a more active way
in terms of what influences processes or
where conditional dependencies are
and so it might be the case that you can
ignore context because you've actually
isolated the system so it's like a
closed system or it's a system where you
can fully describe it
and in those situations context is there
but you've made a special sub problem
where you have all the context but when
you're in a
situation where you don't have all the
context we need to have
a framework for reasoning that has
contextuality by default
so that we can work better in those
spaces
and let's talk a little bit more about
this contextuality by default
cbd not that cbd
this is the authors talking about
contextuality by default just totally
you know
raise your hand or jump in if you have a
thought or a question
and they say that the term um
determining whether a particular

instance of observed behavior of a
complex system
particularly one with memory exhibits
intrinsic contextuality is not
straightforward
so that was interesting makes sense
complex systems
hard to know what to do sometimes
whether you want to include memory and
historiosity or not
and in some ways like if you have like
10 different animals and you think okay
well they're the same
background so we can just consider them
even but they've actually had this
history that differs so they're not the
same
but you want to imagine that they're
replaceable or shuffleable in a
statistical sense but that might not be
and then here's the quantum sense of
contextuality
in quantum systems a set of measurements
exhibits contextuality
if it cannot be characterized by
mathematically consistent
context-free joint probability
distribution
asking whether a set of measurement
outcomes exhibits intrinsic
contextuality
is in this case asking whether a
consistent globally defined
globally connectable joint probability
distribution exists or not
and that's where this formulism of
contextuality by default
addresses questions by labeling context

and includes them explicitly in
conditional probabilities
rendering probability distributions
context dependent
so let's look at the papers
that were cited and referenced and think
a little bit about
where is this contextuality theory
coming from where is it taking us
what do you think oh yeah comment
um that that brought to my mind
um emotion uh related to context because
i've been thinking a lot about um
how like if i'm attached to some idea or
something like that
and then and i just remember that i was
attached to it but then i go to sleep
and whatever and then i forget what it
was
i have a tendency to like really be like
no that was really important and so
regardless of whether it was true or not
i want to go back to it and so there's
i'm thinking about the way in which
emotion is a kind of a context but it
takes you over a longer time
domain because like it doesn't even just
sit in
like a decay rate that would normally be
you know related to the other
information it pulls you
way back into another space so it's this
kind of like
long-range driver i don't know that's
just
context cool
and it brings us to this next question
which is is there

this is the title of a paper that's
cited by um
this offer of 2016 is there
contextuality in
behavioral and social systems so if we
were dealing with this informal
definition of context is just like the
environment it's like
yes the answer is just yes behavioral
and social systems yes there's social
context like oh you missed the context
on this dinner party okay so
off the bat we'll say yes but we're
thinking about it formally and this is
this is the most hyphenated sentence
i've just ever seen in science
so just the number of names
it was quite something to read um
but we're thinking about contextuality
in a more
formal way which is like so let's say
someone says you're missing the dinner
party context
um and then it's like well if you would
have told me all the info then i would
have had all the context
so then that would make it that's it's a
similar
notion but we're going for measurement
systems and um
they're proposing that their cbd allows
one to define and measure contextuality
in all systems
of all these hyphenated types even if
there are context dependent errors in
measurement
or if something in the context directly
interacts with a measurement

these are the two cases that really
matter a lot for scientists
like what if at higher temperatures your
thermometer is biased towards
over estimating or being more error
prone
then you're going to get this
distribution that looks like it has some
sort of super specific relationship but
it's totally not the case it's all
an observation error and the second one
is that the context could directly
interact with the measurement
aka every single time you want to know
something new
so the context which you don't have
because you don't have the full context
because you're going out into the
unknown to make a measurement
is going to interacting interact with
your measurement maybe even modify it
itself
what do you think blue so
i think like you know this just goes
back to that intrinsic
versus extrinsic context like in a
social and behavioral system
there's no extrinsic concept right like
every
er there's no extrinsic context every
like
thing that you're going to try to
measure if you ask like what was the
temperature at that dinner party
or how is the food you can ask like you
know 12 different people
and they're all gonna have different uh
things to say or like did you have fun

or or whatever like i mean everyone you
ask because it's it's
only defined by the intrinsic
contextuality
which is the experience of each agent in
the room
i think cool
and interestingly they look at several
data sets
none of the data provide evidence for
contextuality so again don't just
yes there's a niche there's a context we
know what social context is but we're
actually thinking about it
in this very interesting way where those
data sets don't
have contextuality so a good little gut
check
is if you're thinking about the
applications this paper and you're
thinking about it like social context
it's just not that one because um
it's not and their rule is that
behavioral and social systems are
non-contextual
again if that makes you cringe it's
because you're thinking about it
informally and we're actually thinking
about it in a way where that
claim makes sense that's where we want
to understand
to be like oh right informally we get
where they're saying it
but then we're interpreting their claims
rigorously
i.e all contextual effects in these
contexts or
in these settings result from the

ubiquitous dependence of response
distributions
on elements of context other than the
ones to which the response is presumably
or normatively directed
so let's say you have a driving
simulation and then there's going to be
some
green lights and some big red stop signs
and you do the behavioral measurements
and you find out that conditioned on
whether it was a green light or a stop
sign the people
accelerate or break so that's your
experimental data what you controlled
and what you observed
was stoplight and then whether they
stayed still or accelerated
now that is context um
it's it is context for the person who's
driving that's the social
sense but all the contextual effects
in this experiment result from the
ubiquitous dependence so across the
whole population
of response distributions on elements of
context
other than the ones to which the
response is presumably or normatively
directed
so it's like you thought you were
measuring the response to a green light
or a stop sign
actually you're measuring a social
context
and a group of people who were in a
relation with those symbols
such that they operated in a certain way

so we need contextuality by default so
that we don't fall into like correlation
causation
super fallacious scenarios
here's another case of where context
matters
and what's going to be fun is actually
this decision making is going to matter
for context and arrogancy
here's a paper by value um
local choices rationality and the
contextuality of decision making so
every decision has a context so we know
about that
however rational explanation appears to
be challenged by apparently systematic
irrationality
observed in psychological experiments
especially in the field of judgment and
decision making
so how many times have we heard oh
people are so irrational they'll take
the one dollar today rather than the
dollar 20 cents tomorrow
or they'll pick up their children
from the daycare at the right time
unless there's a monetary penalty and
then that will make them later they're
so irrational
within this one variable that's being
estimated here it's proposed that the
experimental results
require not that rational explanation
should be rejected
but that rational explanation is local
i.e within a context
thus rational models need to be
supplemented with a theory of contextual

shifts

makes a lot of sense um it sounds like something that's relevant to learn a bit and this paper

has probably other relevant things but it's like instead of just saying well who's rational and who's irrational couldn't we just understand that everyone is in a local

logic and a local context and then they're operating a certain way conditioned on that and so it's not a rational versus irrational divide it's like

what's the context who's the agent

what's the niche

what's the context shift so

that's a much more open but also powerful way i think to talk about decision making

okay just raise your hand if you want to pause it

let's take this idea of context so we started with context in the sort of um folk sense and took it to this idea of defining all the relevant variables now let's think about defining relevant variables

in a model in terms of topology

so here's some cool slides from the simmons institute

and in this top right one

it says topology is about distinguishing the continuous from the non-continuous and also about moving around so i was like

that's a cool definition of topology not nodes and edges not how things are

connected

but it's a way it's saying how things
are connected and this slide was really
informative

when we're asking several questions at
once the answers could
obey constraints so like if you're going
to measure the pressure and the volume
and the temperature
of an ideal gas there's an equation $pV = nRT$
that's going to relate those
things and if one of them is out of
whack

like probably something is wrong so
those bundles of answers are related to
each other

could be by the laws of physics like
here's uh you know velocity
and time or whatever it happens to be
like physical constraints
or logical constraints so like if things
are true

then if they're false if they're not
that and then if it's double not
that it's back to being true so you can
have these bundles that are logical
or a bundle that is based upon some of
the constraints of physics

and then also the constraints can be
rows of a table in a relational database
that's the trip because that's where
like sql and a lot of programming comes
into play

and models distinguish good and bad ways
of connecting the dots and bundles
just like continuous sections so this is
kind of like thinking about
sql queries and database queries

in terms of the context

um that's being concerned saying in the
context of this

database i want to make a query within
it

so that's an interesting way to think
about how things are connected and
context but

um this is also stuff that's

we're all just trying to put out there

and hear people's reflections and

thoughts on what do you think sarah i

just wanted to restate it to make sure i

understand it um

i mean basically say they're saying the

the the

the way you move through context sets

creates the topology space

i think that's is that an accurate way

to phrase it

um just another

aspect of it yes like there's certain

in the space of making multiple

measurements

those measurements can have different

topologies like if you make three

measurements they could be in a line

they could be in a triangle like there's

that's not at the formality that's on

here but when you're making different

measurements

they can either be in a well-behaved

bundle or

a not well-behaved bundle and then watch

whether it's well-behaved or not

is um

related to whether it blew what do you

think

oh yeah i mean it's a hard thing let me
just i'll jump in for you
so so um going back to like the
intrinsic and extrinsic um contextuality
like so it's contextu it's contextuality
when you have like a set of measurements
that you know um are
corresponding measurements right like so
i mean it's like um
i don't know do you have bio background
sarah like like i'll i'll talk about it
just in terms of like
you know rna sequencing right like you
can sequence your sample
in a different lane on the same flow
cell or in two different flow cells
and there's variability right so you
always have to try to like design
controls into your experiment from flow
cell to flow cell from lane to lane and
so that's why you replicate
the sample sequencing just to kind of
control that that
variation in just running it at a
different time the temperature in the
room might be different
or you know the machine might not be
feeling as well that day or
or whatever so so it's these kinds of
are the measurements
are they all like do they all link up
together
like is there like a correlation between
all of the measurements or not
so that's the well-behaved bundle is
when they do all link up
they have that shared um information
right so also

i think as we get into the paper maybe
it might start to make more sense
yeah okay let's go to channel theory
okay so this part was super fascinating
um and i'm drawing a little bit on one
of the background papers mosaic of two
spaces and channel theory one
by fields and glazebrook which we're
going to talk more about the two
mosaic papers soon but let's
contextualize this in terms of
information theory so-called info theory
but i personally call it dis info theory
and i've called that for a long time and
now i know why actually
so here's the second paragraph of claude
shannon's 1948
work shannon writes the fundamental
problem of communication
is that of reproducing at one point
either exactly or approximately a
message selected at another point
wait is that the problem of
communication
am i trying to reconstruct something bit
for bit
via communication or is the goal of
communication to
impel joint action to update
other agents not to replicate a message
shannon's working with telegraphs
and copper wire so it makes sense why
you want high fidelity data transmission
a signal of transportation of
information but
let's think about how broad people have
gone with info theory
according to shannon um and then he

writes

frequently the messages have meanings

they refer to or are correlated

according to some system with certain

physical or conceptual entities

these semantic aspects of communications

are irrelevant to the engineering

problem

o semantics are relevant to

communicating information semantically

for people or for what the significant

aspect is that the actual message is one

selected from a set of all possible

messages the system must be designed to

operate for each possible selection

not just the one which will actually be

chosen since this is unknown at the time

of design

so fair enough a through z you want to

have all of those options because

you don't know what next letter needs to

be poked

but i think if people read through

shannon's work

and considered a few of the aspects of

the shannon entropy like how it

disorders

messages like it just asks what's the

symbol frequency of the whole message or

page or genome

if you think about how it removes the

ordering of

symbols and is actually asking about

replicating bit for bit messages

not communicating in any sense people

would really consider what information

theory is

thankfully in this paper the

fields of glazebrook wrote about
comparing and this was cited in the
2020 paper comparing and combining the
shannon theory of information with
channel theory

okay so um though very general as a
quantitative theory of communication
flow

the original shannon theory had largely
overlooked the question of semantic
content

in any dretske type theory the basis of
semantic content is in the world i.e the
events or situations that signals or
states carry information about

the car you know it's out there it's in
the world by showing how local logics
are connected by information networks

channel theory provides a generative
general qualitative theory of
information flow in this context

so that's pretty interesting it's more
about

information flow that semantic rather
than syntactic

and then this last part here um

this is there's more to say here but all
while in 2004 create a synthesis of
shannon's quantitative information
theory

with the bar wise seligman qualitative
theory

to address how the question of how
specific objects situations and events
carry information about each other
that's what people are always talking
about when they want to be talking about
sharing information

not sharing surprising symbols but
updating each other
semantically informationally in the
terms of bayesian priors
about the world things that we're
referencing in the world the temperature
not just as a variable but like as about
something in the world
so some thought questions is
shannon 1948 correct about
the stance taken on the fundamental
issues of communication or what
communication is or isn't
how is channel theory distinct from
information theory so
called this info theory we can start
calling it and then how is
real communication the kind that we care
about and want to model in many cases
how is it both syntactic and semantic
all these things at once
grammatical logical as well as lexical
word-based it's also narrative driven
a narrative or the narrative context can
make just one word
semantically powerful speak now or
forever hold your peace it's a narrative
context
where one word cannot be reduced to just
the bits
or something like that oh well but no is
a common word or yes it's a common word
it doesn't matter that it's a common
token because it's semantic information
and then
how is it contextual blue
so i mean i think that there's this of
like

i i like shannon information theory i
like how it's very
logical and it appeals to like my my
my brain structure um the
semantic information that's contained in
this channel theory
description is very squishy uh and and i
think maybe
um not as appealing to me but i
do think that it's more correct right
like when you say when you reproduce
the information like if i say
um something like oh it's raining
outside i live in the desert so like
it's raining outside it's like a really
awesome like event like let's go play in
the rain right
so when i say that and someone else says
oh oh it's raining outside
like like oh i'm like my plans are you
know destroyed because i wanted to go to
the park and i can't because it's
raining outside so
so it's two it's the same information
but
like i the way that i said it was very
different in those different contexts
right
so cool let's return the discussion
from information flows and communicating
information and we're kind of
thinking shannon syntax information
reproducing bits
bits on this computer to bits here we're
going to take it up a level to semantic
information
and then semantic information so like
i'm thinking of a purple cat

now you're thinking of a purple cat but
that could have been in a different
language and it's not really related to
how surprising each
word is let's return to this question of
local logic and contextuality
so here's um a paper not sure if it was
cited or not
but it's going to bring together local
logics contextuality and
topological approach so we're just going
to kind of give multiple codes of paint
on everything
local logic is like the local rules or
the context
and in this um kishida 2016 paper
they wrote the topological approach
characterizes contextuality as
global inconsistency coupled with local
consistency so that was kind of cool to
me it's like if you have this field
and it's globally consistent then it's
like transparent there's there's nothing
inconsistent to speak of there's no
fracture in the glass
so it just is purely transparent so
um that's just the same it's part of the
same context
but local consistency coupled to global
inconsistency
is like a bubble it's like it's an
inconsistency in the tank of water
but then locally it has another
transparency
that makes sense in that local context
and so you could have like a computer
program and then that could be a local
logic but it's embedded in another

logic of being a turing complete or you
know whatever kind of computer
and then within that program there could
be like a simulation of another program
of an n64
running another program and even if it's
all the way back up at the top it's
running on linux
but that local program is running on the
n64s
local logic and then another kind of
metaphor
oh and then in the paper they write our
goal is to capture this local
consistency part
not just the global inconsistencies the
global inconsistency is like our
receptor is not getting any photons
boom it got a photon inconsistency so
not just that part
but the local consistency part which so
how do we know
how do we know that it was consistent
before that photon hit which requires a
novel approach to logic
sensitive to the topology of contexts to
achieve this we formulate a logic of
local inference by using
context-sensitive theories and models in
regular categories
and that's going to be powerful here's
how i was thinking about this in terms
of chess
so the rules of the game of chess are
like
which piece can move which way and then
some of the imperatives like
you have to um alleviate check otherwise

you lose the game
that's the context the local rules of
the game
then there are local or situational
patterns like a pin
where one piece is like in front of the
king and it can't move out of the way
because then the king would go into
check
so that's not a rule of chess but
it's an emergent pattern that arises
from constraints in the space
given the local logic of chess but now
we're in this even
sub-local logic i'm not going hyper
formal maybe chess players have another
way of doing this but it's like
there's a local logic in the corner of
the board with a king
and not every piece or rule of chess
like okay well the pawn can move two or
one
but that rule might not matter for this
pin with these pieces
and so not every single rule or piece
is going to matter for these local
logics
and maybe there's a way to pursue this
topologically in terms of the
connectedness of different relationships
of the system
let's take that to choose spaces
okay so this one was um you know it's
the second section of the paper so it's
like okay you gotta
understand and and learn this one so we
were kind of excited to do this
so they start by defining one of the

simplest categories so we're in category theory

is the category of two spaces which we're going to talk more about and it basically is about a set k and then within k there are kind of two kinds of things objects and attributes and the two space is like a matrix that is the relationship

over that set k of the relationships of a

and x it's so it's actually probably a data form that we've seen a ton of times like

here's ten people and then have they RSVP'd for my event or not

yes yes yes no one one one zero

that's a two space for RSVPs but it's like that

you know here but it could be a lot more general and we're going to keep on generalizing on that but the basic two space

is a matrix but it probably can be higher dimensional

that is just a regular matrix but

a topological space is also a matrix so

you can have like the adjacency matrix

right like all the nodes by all the

nodes one or zero connected or not or a weighted graph

how strong is the connection or it could be symmetric

and undirected or there could be a from

and a two so like all of topology

is matrix forms and it turns out that

they're two spaces

but also we can think about like
programming languages
and we can have a type as
one of like an attribute or a type you
know is the wood
tabling shoe space or is the object a
table
that's a chew space so there's a lot of
programming relationships that the chew
space covers
probability spaces um coin flip you know
content you know that is it gonna happen
or not
so this it's like if it's a matrix i
don't know if it's you can go as far as
say it is a two space but it
certainly a broad family of
objects or categories i don't even know
what to say there are within this
two-space
paradigm there's a few more definitions
because we looked up just a bunch
here's from wikipedia and basically the
big idea
is that the true morphisms are
transformations on
two spaces and if things are set up
properly
the morphism's continuous and the pair
of functions
has some special relationships so um
one example this makes me think of is
like the typographical number theory
in girdle escherbach where it's like one
plus one equals two
and then you make a set of text rules
like you can add an s
to the string under these contexts or

you can add um or you can you know
it's like it's basically doing addition
and subtraction
by typographical like letter based rules
so if you can prove something in one
system
because you've super strongly shown that
there's a morphism
you can show that that same relationship
should exist in another system
it also turns out that's actually at the
heart of like the whole gerdell
question but that's the static
interpretation
and then um here's dynamically two
spaces transform in the manner of
topological spaces
um yeah don't go too deep down here but
there's special relationships of two two
spaces that are chevmorphisms
and then if that morphism is
bi-directional it just allows a really
nice thing where you can sort of make a
finding on one side and then arbitrage
it
to some relationship that might have
been difficult to show on the other side
so it's like a perfect language mapping
you could say you know
the the boy is younger than the car if
you had a perfect language mapping
semantically you could say it in another
language and the semantic information
would be perfectly preserved
so then maybe one person has a piece of
semantic information in english and then
they could convey it through the perfect
morphism

and it would be correctly semantically understood on the other side okay here it's just a couple these we just we wanted to actually understand it and also um shout out to master students everywhere so here's history from this paper the chew construction takes a symmetric monoidal closed category $\mathcal{V}$ and with pullbacks i got reading it and an object k of $\mathcal{E}$ and completes $\mathcal{V}$ to a self dual category $\text{Chu } \mathcal{V} k$ the details of this construction appear in ph choice master's thesis published as an appendix to his advisors mbar's book introducing the notion of a asterisk autonomous category so chapter that was an appendix and now here we are many years later so it's all good to research whatever you're researching on if it appears to you to be the big question here's barr's perspective so bar is another perspective on shoe spaces barr had this idea during his sabbatical to study duality in categories in some depth just what most of us take our sabbatical wondering he wasn't wondering about the simple dualities but self-dual categories like complete semi-lattices or finite abelian groups he was interested in the possibility of having a category that was not only

self-dual

but one that had an internal home and
for which the duality was implemented as
the internal home

into a dualizing object it's technical

so duality

is going to mean something different to
a mathematician than a non-mathematician

and

i don't know what a home is however

searching for an internal home i just

thought okay the dual it's like forwards

and backwards or it's like it's

it goes both ways that would be nice you

know reversible

it's like those are always good things

and then the internal home

internal homes when chained together

form a language called the internal

language of the category and i thought

okay so now we're going from categories

and attributes to like

something else the most famous of these

internal homes

are simply typed lambda calculus

which is the internal language of

cartesian closed categories and the

linear type system the internal language

of closed symmetric monoidal categories

lambda calculus is something that's

quite broad and central to

computation so this is like something

about how logic

local logics and nested logics

are related to categories in an

analytical way

such that the kinds of logic that we see

on computers

is only a type of it because in a way
we're dealing with syntax based logic
you know you have two files are they the
same it lines up the bits and if they're
the same
it's yes no but what
would this mean if it could be
semantically
the same and how would we make that
non-squishy because
to ask whether semantics were the same
you need all the context
so that's kind of this fun twist and it
makes sense
why for simplicity you use frequencies
statistics instead of bayesian
inference because it is hard to do
sometimes
or you do have to grapple with unknown
unknowns and it's easier to just do a
t-test
or a multi-linear regression rather than
some other type of generative model
because that takes like bravery to
specify your uncertainty
and to give support to why you did a
certain thing rather than just say
well you know the countries differ with
a p-value of 0.01 but
it's easy just to say that and move on
rather than actually do some sort of
rich modeling underneath
okay true spaces and channel theories so
like we're kind of
just trying to bring all these two you
know connect two of the ideas and then
you can start nucleating some ideas
from the ones that are making sense in

channel theory which again we're thinking of like the um more general cousin of shannon dis info theory in channel theory

two transforms these are the transformations between the two sets um or spaces become infomorphisms which are natural maps between classifiers

okay so blue what would you say about the red underlined part or what you wrote at the bottom um so this will i'll like transition you into the

next slide maybe here in a minute but um just the red underlines part it's the transform from a to b so if you have one two space and another two space the whatever it's like you know you can think about like a territory and the map of that territory like whatever transform going from the territory to the map that is that channel right is thought informally of as a channel um that is uh like the channel that we're talking about here in in terms of channel theory and so these um the the classifiers here so the um a classifier is the um set of where did i have that here somewhere classifiers link tokens to types oh yeah but so oh that's that's this is the definition of a classifier so classifiers link

tokens to types

that encompass them oh yeah because
you've already read this so the channel
theory the

two transforms become infomorphisms

which map between classifiers

so if you have different sets of

classifiers right so

classifiers link tokens to types that
encompass them

so for the type stoplight the set of
tokens could be like you could have a
red stop light a yellow stop light or a
green stop light and that's also known
as a local logic the classifier which we
talked about local logics

just a few um slides back but flip to
the next slide and i'll continue this

so mapping between classifiers

so here if you have the classifier

stoplight and then you have like

what the motorist should do with the
stoplight so that's the second

classifier you could have the tokens
like

stop at the red light go forward at the
green light or proceed slowly at the
yellow light or

like get out of the intersection or

whatever it is you think you should
actually do at the

at the um stop light but then you can
have other classifiers too so it doesn't
just have to be the stop light and the
what the motorist should do

it's also like what pedestrians should
do and like at the game like red light
green light that you play when you're a

kid like you should run forward run
slowly or walk or or then
and stop so i mean there's different um
classifiers that you can have that all
mapped to the same central information
core right and so that's what you see
here the classifiers are
these things a_1 a_2 and you can have many
of them up to a to the k
and then you have at a core this
information
channel right so this is the
infomorphism that allows semantic
information not just shannon information
to be transmitted between
classifiers so this comes down to having
a shared memory so we all learn from the
time that we're six
that red means to stop green means to go
yellow means to slow down so so but to a
six-year-old it means running
you know when you're a pedestrian you're
looking out for other cars and and there
are other behaviors that you do
and as a motorist also and as a police
officer policing the motorists there's
all kinds of people that have different
reactions to what
the stoplight is doing but that's all
based on this shared
memory like we've everybody has been
taught like this is what you're supposed
to do
out of light um and so so that's this
core
information channel um through which
information
flows it's can be also be thought of as

a shared memory or
um a shared context if you will
cool one thing that that makes me think
of here
is like this sentence that said um uh
the existence of a true transform a to b
is equivalent for every flow formula
valid in a
being valid in b so it's kind of like
you have the chessboard
and the board position says it's game
over and then you have the
the data file and then those two are
within
a bigger c that is establishing
a connection between these very
different types and so it's like there's
um in the example you gave the local
logic
is like there is a total logic that it
does all come back to
even though there's disparate types and
there's sub logics
let's hear what everyone else and chris
can bring to this discussion because
it's like super interesting to think
about
um yep okay so what about sequence blue
um so the sequence here uh
i just wanted to put this in because i
thought it was it was
important for information flow so
sequence uh
when a secret incl encodes a semantic
like a causal constraint
uh that effectively functions as a logic
gate so this really what it's what
enables

semantic um information as opposed to

just

shannon information so it's like you

have the message but there's a logic

gate there

that says if the message is applicable

or not applicable or to be transmitted

or not

um and also that it um i think

puts forward into the um it enables

putting forward into the

scale-free or scalable uh infrastructure

here i think

um and then i like all this like math

symbols like i've never seen this stuff

before these like

what is this like parallel line lollipop

stick i don't know

um but but i just i i put a box around

this because it's like oh

that goes back to bay's probability all

right i know this

so it's coming home again

cool i think this one just reading this

one um x

then shape n to assign a probability x

satisfies

n so maybe it's the sign of satisfaction

um

i guess it just says something that we

don't know what it looks like

um but it's like there's a satisfaction

of

x to assign a probability that x

satisfies n given that it satisfies

m so it's like what's the probability

that we're in checkmate on the board

given that the file says checkmate well

if you have the morphism
it's perfect but if you don't have that
morphism
then it's probabilistic or it's
different
so it's a conditional probability that's
conditioned upon the way that you can
map across different kinds of things
and it was the perfect connection to
beijing inference
because at the bottom here we have u_m
basically m
satisfying u_m a
is the true space that's the the scarlet
letter the fancy a
is the two space u_m and so
this is almost like saying m is
satisfying some true flow
relationship with n that
is itself about how tightly the spaces
are connected
and so then we can take this $p(m|n)$
single line
not satisfied u_m and it is
belief updating and so then
uh yeah it provides the necessary
semantic consistency for such updating
infomorphisms thus capture a significant
representation of bayesian inference
with the necessary coherence since the
target classifier context emits the same
semantics as that as the source within
the information flow
information or infomorphisms are
information flows
that capture the updating of something
semantic because it's semantic in that
dretzki

meaning is in the world way like the
prior and the posterior are
about the world and then as
new data or events come in which i
didn't think we captured as much here
but like this event theory connection
it's like bayesian um statistics is kind
of like
the steady state and then new data comes
in it's like an event
in a network and the in that event
propagates through a network
of semantic information flow and then
that updates the system and maybe it's
like the system's already vibrating and
so the new event comes in
nothing changes with a vibration that
might be a situation
like the system semantically is already
at its non-equilibrium steady state and
so it's like not
surprising but then there's other kinds
of events
that enter a system that might be more
surprising
but it's all about the semantic
nature of a system okay so wait wait yep
can i just add on to this yep but like
in our in our
um contextualizing like and i don't know
if you were going there or not but i
just want to explicitly state that like
bayesian inference is the event
like which we are updating with active
inference and
category theory and two spaces were you
like like
implying that or i just wanted to state

that explicitly because
in this context contextuality talk about
contextuality
like we are updating our own like events
with our with our own knowledge
yep and i i think what's sort of fun
about this
semantic info flow idea the whole
channel theory thing which i really
didn't
get exposed to in a lot of contexts was
that
instead of the propagation of bits and
bytes on wires
and the signal fidelity and then it's
like well then where does the semantics
get unpacked
does it get unpacked at every little
post office and they just say no no just
syntax all the way down
so it's inherently reductionist because
you go no we've explained away
meaning because we're studying the
connection
from a entropy perspective and then it's
sort of a strong
internally logical argument which is why
there's people who believe
it and then it's like could we actually
think of semantics as the heart of the
question
because when we're talking about
communication of action or modification
of the niche
or solving the frame problem we don't
need the syntactic differences we want
the semantic differences
like what has changed my car is broken i

don't need to know the syntax
but from an action-oriented semantic view
that's what we're communicating on and
then we're seeing these specialized
communication
message passing systems like on a
silicone chip
it's a special case it's a controlled
system
where we've defined the context
semantically
so that it isomorphically maps onto bits
and voltages okay
here's the two fields and glaze brook
papers from
uh 2018 uh
mosaic of two spaces in channel theory
one and two so
one category theory concepts and tools
is like a lot of
formalisms diagrams but two has some
nice images
and that's applications to object
identification and
neurological complexity is
parts and whole relationships and
this is to go from the two spaces to
this
cocoon a system capable of both object
history construction and its
dual category learning
so just there history reconstruction and
its dual category learning it's like
i'm learning about cars being broken or
not
and then if you tell me it's a broken
car i can generate a vision in my mind
of a broken car

so it's like you have a bi-directional relationship that's that's open to generating specifics from the abstract or updating the abstract from the specifics which is really related to this hierarchical bayesian learning and active inference and they have reached after many years of research and understanding that objects that have this sort of analytical relationship between object history construction and category learning with a singular category of maintenance as a special case not sure i think that maybe that has to be one object that does both a self dual not just two objects one that does one part one that is the other is characterized by a cone cocom diagram a cone cocon diagram captures the simultaneous upward and downward flow of constraints that characterize human visual object identification and it is reasonable to suppose other sense modalities and they then take it very instantly into something fascinating which is the functional duality between a high precision expectations and high precision inputs in an active inference system and hence the duality between dorsal active and ventral passive attention systems

so it is about this precision modulation
and the way that you get high precision
expectations and high precision inputs
so how do you have a clear generative
model of the visual field but then at
the same time
you want to be able to recognize new
objects at high fidelity as they come
into the screen come into your field so
how are you going to have high precision
generating and high precision learning
and there's going to be this rich
multi-level
um ongoing integration of priors and new
events and posteriors
that's semantic so it's like i'm looking
at oh it's a face and then my
eyes engage in an action selection
pattern appropriate for a face
that reveals some sort of anomalous
information pattern
and all of a sudden the meaning has
changed and so it's almost
like what is crossing the interface
perhaps is
the the data the observable the
measurements
but what is crossing
is actually like the reconstruction of
semantics
rather like if it's a car driving across
your view but it's going 5000 miles an
hour and you can't see it
then for you as the measure it doesn't
matter because you didn't see it
so even though it was perhaps there but
it's just a blur
and so this is just getting at something

very interesting which is these systems
that are exquisitely generative yet also
hyper able to recognize and
to update the generator based upon
the specifics of which have been
generated oh all cats have stripes
all right you just saw one that had
polka dots and now you can generate that
and then maybe you might generalize and
even generate another type of pattern
of cats and then you could see whether
that was

accurate or not so here is the diagram
that blue talked us through
here with the core information channel c
at the top
and so this is the cone diagram on the
top it's it's like a pyramid diagram but
like

maybe if it rotates it's like a cone or
there's some other way of thinking of it
like a cone

and you have the classifiers and this is
the whole choose space as classifiers
scarlet letter a

choosebase so the classifiers are
linked through infomorphisms semantic
bridges

to the core information channel so like
here's the chess board here's um another
two space

and here's another two space for chess
and they're all linking to this like
game of chess

local logic and then there's a map
between the board and the computer
program

so g one two two three are the

commutivity so this is like
you got a link between the board and the
computer and the computer and the
drawing
and they're all infomorphisms to see
the game of chess so semantically if
there's a checkmate on one board it's
like a checkmate across paradigms
now when you're um
when this is a a tight cone and it
represents like a huge variety of things
and then i'll get you the second blue
the co cone
is when we oh yeah go ahead go ahead
good let me let me interject before you
do the cocoon that's why i was like
okay so i want to interject here for
sarah cause remember we were talking
about like the
the observables being like a tight like
a tight little
cluster of of measurements versus a not
tight cluster of measurements
and so here they define it in this
before we go into the cocoon
um so so you can kind of take this back
to like
the stop light and the kids playing and
what the driver does what pedestrian
does and whatever but
but here if you think about it instead
of like that the categories or the sorry
the classifiers
uh it says when we define a finite
information channel
c uh as a finite indexed family
of infomorphisms those are all the f's
having a common core the component

classifiers will be taken to
characterize
a system of observables or some
subcomponents thereof
so here they're specifically stating
that all of the classifiers here
are all of the things that can be
measured or observed
in a in a context that's extrinsic as
opposed to
intrinsic i think so i could be wrong i
hope that someone someone corrects me
but i feel like this is where
where it's going
cool cone diagram then we introduce
the co cone the shadow as above so below
wouldn't have it any other way a
commuting finite
cone cocoon diagram which one's the cone
which one's the cochrone right
comprises both a cone and a cocoon
on a finite set of classifiers so
they're mapping to the same
set of a classifiers the same two spaces
are being bridged
and um then you have like sort of two
systems
looking at the same bridge
and then uh if there's a cumulativity
across the elements of the of the two
spaces
we have that from above and then there's
infomorphisms from all the a's to c
prime
and all the a's um to d prime
then it's a special kind of object what
is special about it it captures
in addition to a ton of other stuff

probably a more subtle duality
remember the whole thing about studying
objects that were dual
objects from like bars 1975 paper and
chris fields other research like what is
a dual
object shadow a more subtle duality
between processes so not an object
duality but a process duality maybe
forward and backwards
it enables object files object tokens
and object histories to be viewed not as
tokens but as
types that organize respectively
trajectory components that's files
features
that's tokens and feature based singular
categories
histories into mutually consistent
collections
so this is some next level token algebra
and i wondered if it related to crypto
because it's like instead of the history
like the object history my web history
well that's actually a feat that's a
singular
category it's a singular trajectory if
you take this list of a million things
and it's like one of a unique kind of
thing
and then even objects and tokens and
descriptors
are all understand under being
understood as highly relational
highly semantically mappable way
so that's pretty cool and i guess we'll
look forward to seeing what the authors
can help us with or like what system

could we apply this to first
or what would be an empirical system
where this is being measured
or what would we gain by measuring
something this way
yeah what are various examples of
context um
and then here's just a few figures from
that other paper the number
mosaic two so here's a cat well it's an
image
it's a pattern on a google slide who's
to say it's a cat right that's semantic
but there's all kinds of things that we
might want to
have so it says an object token is
classified
at construction into multiple types by
distinct but cross-modulating processes
this includes animal animacy and agency
detection emotion-mediated threat
detection and entry level
followed by super and subordinate
categorization into types of objects
so it's like depending on the context
and the agent
observing this stimuli whether it's a
predator or prey
whether it's a cat or whether it's felis
domesticus
or whether it's a sub breed of cat
whether it's something that you approach
or something that you withdraw from or
something you associate culturally with
good or bad luck
that is like the agent's ontology
semantically
as they interface with a cue in the

environment

and then this is a funny image it's

mapping chris fields brain

so it's like it's it's it says like

cf that's that's him so here's cf

is a human humans has a brain

and then here's like another ontology of

brains a private brain is a mammal brain

is a vertebrate brain is a brain

so you can go superordinate and

subordinate

categories in a relational ontology

as well as have these like totally

disparate classification

schemes just like you could in

programming so you could say this person

has a height and a weight and that's in

their public health

record and then there's this other thing

are they a friend or foe and that's my

private information

and then all of those are being linked

semantically

in a special kind of system all right

that kind of that's one little section

here is bringing in this element of

concurrency so some people will have

heard about this from

programming languages like go lang or

from other kinds of concurrent

programming

but this is pretty interesting because

it's from 2005

and it's called paper by pratt it's i

think cited in glazebrook

and fields two spaces and their

interpretation as concurrent objects

so it's yet another way that we can

think about two spaces not just as
mapping
attributes to types or doing all these
other transformations but like
also as concurrent programs so
they write here that the two pressing
questions for computer science in
2005 maybe forever are what is
concurrency and what is an object
the first question is of interest to
today's sibling growth industries of
parallel computing and networks
both of which stretch our extent models
of computation all well beyond their
sequential origins
so you have the turing tape and it's
just zooming backwards and forwards on a
tape it's a standalone machine it's not
connected to the internet
when you have machines that are in
networks then
there's semantic logics that are
happening due to those local logics that
are boolean
but at a higher level they might be more
like a complex adaptive system
like a ddos attack on a network
or a certain type of governance game
theory
uh situation in a crypto economic
network using a cad cad model
those kinds of complex adaptive dynamics
at the network level may or may not
actually um be able to recapitulate
from the syntactic or from the lower
level
and so that's the first question is how
do we think about

um about concurrency and about designing
for
good concurrent systems when our
parallel and distributed computational
and even unconventional computational
paradigms are
rapidly expanding the second question is
relevant to programming languages where
the definition of object
seems to be based more on whatever
software engineering methodologies
happen to be in vogue
than on the intuitive sense of object so
in other words people use
object and process in a way that might
have a name space meaning for computer
science
but is actually less connected than we
might want it to be
to the formal math and this is pretty
another last quote on this paper they
write in as the separate chapters of a
theory of concurrency
all these different topics petri nets
event structures and domains stone
duality and process algebras
these topics make concurrency something
of a jigsaw puzzle
one is then naturally led to ask whether
the pieces of this puzzle can be
arranged in some recognizable order
and then they give a strong endorsement
of the two-space framework
so it's just interesting that these two
spaces can be objects
of relations that can be processes but
also
it can include context that's what this

paper is about and
algorithmic and game-like elements so if
you can do it in goLang
if you can do it in the process algebra
if you can use higher order reflexive
process algebra
maybe that's a two-space all right
let's talk a little bit about ergodicity
so this is going to have a few
slides and definitions so on the top
left
a process is said to be ergodic if the
ensemble average
and time average are equivalent and
here's a more formal definition from
this is from the neurobytes
site a more formal definition is from
ole peters who's uh
been at the santa fe institute an
economics professor
the ergodic hypothesis is a key
analytical device of equilibrium
statistical mechanics
it underlies the assumption that the
time average and the expectation value
of an observable are the same
where it is valid dynamical descriptions
can often be replaced with much simpler
probabilistic ones
time is essentially eliminated from the
model so
this is in on the website neurobytes
it's a
live and updating but here it's just
static it's kind of
thinking of two ways you can make purple
from blue and red
one is to have the spatial uh

resolution increasing and pretty soon
semantically it's interpreted as purple
so if you doubt that you can zoom in on
an old comic book and see how they made
color

which is by increasing the scale of
resolution really small
then it makes it look like a new color
is there or you can flash things really
fast and then it will appear
as if there's purple and those
equivalencies

or ways of reaching and also deviations
from ergodicity spatially and deviations
from ergodicity temporally
are what we're talking about now why
does the ergodic assumption matter and
what happens if this is empirically
if this is violated so here's an example
from modern portfolio theory
which is what ole peters and others like
glenn begerman

study which is that and this is quote
from neurobits

modern portfolio theory stipulates there
exists some optimal risk to return
profile

but the theory rests on the ergodic
hypothesis by conflating an
ensemble aggregate return with the
individual's path-dependent return
through time

so it's really interesting um and this
is related to the st petersburg paradox
and a couple other economic ideas
that basically the total population's
performance or
a hundred different portfolios

performance at one snapshot is not the same thing as any portfolio being carried forward through time why does that matter how off on the wrong track would we be so here's some funny examples of ergodicity violations and also a hypothetical example of ergodicity so um number two from the neurobiters if it takes one pregnant woman nine months to give birth nine pregnant women should only take one month and then this says recipe says bake for 60 minutes at 200 degrees but i'll just bake for 30 minutes at 400 degrees so like if the degree minute is all things being equal why couldn't we just vaporize it with 10 000 degrees celsius for just a split second um it turns out that there's a process that has to play out and so yeah what do you think about this blue or how would you link it to anything else we talked about today um i'm like i don't know i think my brain is is uh i i'm still giggling at the nine nine segment uh months i'll say something i i find those examples actually worse than just giving me no example because like ergodicity implies you have to have a system level demarcation and so you know to compare it to like

nine pregnant women it's like well
the the system is the woman so you can't
really do that in the analytic it
actually makes it harder to understand
which is good good point yeah it's it's
it's all right
it's like but it just it yeah i think it
makes it
more confusing yeah i agree it's not a
perfect
mapping it's like syntactically it's the
kind of error you'd be making like
here's the ergodic in a lifestyle span
but then
if you go too down the rabbit hole we're
trying to add the variables that's why
we always pull back to the actual
formalisms so here is from math
encyclopedia
and here it's i thought it was quite
interesting so
it's about trajectories of points and
saying that as you'll hear with many
other definitions that it's about
visiting a phase space through space and
through time
that was sort of that first example
here's what was interesting though
when the birkhoff ergodic theorem had
been proved it became clear that
ergodicity is equivalent to metric
transitivity measurement transitivity
the more general situation it was no
longer suitable to talk of the equality
of time and space averages
systems with an infinite so what kinds
of systems
are beyond spatial and temporal

ergodicity

systems with an infinite invariant or

quasi-invariant measurement

not only flows in cascades but also more

general transformation groups and semi

groups

transformations groups categories semi

groups we saw abelian semi groups

earlier

in the bar paper so it's almost like

we're taking ergodicity even beyond

the trajectory visitation idea i don't

know where it's going

in the way that the authors are using it

but it's like there's there's like the

you know the totally inadequate metaphor

like of just making a total category

error with

the recipe and then there's the more

empirically observable strategy

violations and like you're going to have

a bad strategy if you make if you're

using a model with an ergotic assumption

you're gonna do strategy that's just

ridiculous you're gonna fit

what seems to be the best or the most

likely strategy and you're just gonna be

way off base

empirically but this is even taking it

to another way okay let's just

with a couple of these questions bring

some of our clothes

um i think you may have raised this one

sarah but it was asking that

this context by default

generalizes the intrinsic contextuality

of quantum theory

to the case in which other properties of

a context for example the order in which
questions are asked
also affect the distribution of a
variable of interest

this kind of direct influence is
generally avoided by experiment design
particularly by imposing no signaling
conditions

in physics but it's virtually
inescapable when studying complex
macroscopic systems

so these are some pretty open questions

how does context relate to ergodicity
especially when systems are changing
their context or

vice versa or both and

how uh does this contextuality by
default

specifically help us here and then how
does fields in glazebrook build upon
contextuality by default

to maybe add another plank to this
question okay

here's another set of questions the
existence of complementary observables
has perplexed physicists for decades
feynman characterized it as the only
mystery of quantum theory

how are such paradoxes resolved at
the implementation level

if they are or worked around if they're
not what happens when workaround
mechanisms fail

what is required to detect failure post
hoc

ability to prove this experimentally
would begin to answer the broader
question of how prior probabilities are

represented in memory and how different
sets of priors can be deployed in
different contexts
so how do we know what we don't know for
a given context and how can a context be
made to entail
unknown unknowns how do we bound our
not knowing and i think part of the way
is going to be
defining everything as the markov
blanket
which is going to be us internal states
and blanket states is what we can
observe and control
and so everything is going to be
conditioned on generative model of
external states
so we're never even going to peep
outside of the blanket states
we're going to be super serious about
modeling ourself
and the blanket states as interacting
with and modeling external world states
so instead of how tall is the tree
you're going to have like you
and the measuring device and the
measurement in a system
and maybe for some easy questions it's
going to be total overkill
but i think for other questions it's
going to really open the door
because once people frame it in this way
what was an
unknown unknown is going to be
something that we can actually
encapsulate
so that's bringing us this question
here's a quote

we apply this development of ideas to
recover among other things a new
theoretical and inferential role for the
markov blanket concept we read this in
the introduction because it's a key
point
as the locus at which free energy is
minimized and
the epistemic knowledge barrier that
keeps contextual information hidden from
the observer
so where is markov blanket coming into
play
what does this have to do with our
active inference
modeling and learning and then just all
these issues that we've been talking
about
representation internal versus external
perspectives and um
where is this paper coming into play
okay then sarah this is one of your
suggestions slides from a couple weeks
ago
so but it's relevant here again so
what maybe what if you want to anything
that's interesting
like about context or frame or reservoir
just you're you know a high foraging
philosopher of science
so it's always good to hear what you
think about it and my brain is pretty
fried but i
i think i can um my eyeballs are fried
anyway um the thing that
strikes me as a kind of contrast about
what i'm able to read from this paper um
is i i don't know i get the sense when

they're writing in the paper that they
just have this like you know how can we
have the eye of god in like all contexts
and shift
like i you know or at least they're
talking about that in the context of
like modeling and talking about how our
world actually is but
um the thing that makes reservoir
computing so powerful
one of the things is is its
non-ergodicity it's it's kind of natural
partitioning it's it's um
heterogeneity in some way um and so
i don't know that's really interesting
to me this kind of how can a body
compute
um and what are the attributes about a
body that
what are the kind of imperfections about
the body that make it
um that make it intelligent in that
sense
so so yeah you know
in what way do channels form like these
natural cutic natural joints or make
these kind of boundaries and
and why i don't know that's those are
the interesting parallels for me
yep that idea of like it's the the local
consistency the organismality the ending
of itself that's like the kantian
organism concept but it's an aberration
it's far from equilibrium from the
global
context which is just hydrogen molecules
dispersed
so it's like when you have a global

disordering or some sort of perturbation
and then you have a locally organized
bubble
it's then it's like it is our unique
differences and situation that allow
that agent to be
adept in the niche it almost couldn't be
another way or it's hard to see i don't
know the thing that's interesting
is like you don't even have to think
about well
i'm sure there is more you have to think
about in terms of adaptation and you
know response and things like this but
just
you know how does a rock compute and and
there was one paper that i was reading
where they were talking about like well
okay if you stick a rock in the oven you
know it doesn't necessarily heat up in
all the same places so you do have this
kind of um you do have these
these um non-homogeneous kind of
attributes about the rock and
and it's a it's a very loose analogy but
you know
that is essentially like this this um
non-boltzmann type
type distribution that is what makes
these bodies computable in some way that
is
yet still a little bit mysterious yep
and it's like if it were um a little
worm you said if it oh its body
is set up with the muscle the connective
tissue in the nerve so that it moves
away from
a prick from on the side it doesn't need

to be computing where
everything is if it's body is able to
respond
then it can do it needs to do in the
niche okay
but i mean you don't even need to be
alive that's the thing that kind of
blows my mind about how this computing
works
yes so
implications next steps the first so
this is
good that they went there so the first
question which is expanded upon
more in this bestiava paper is exploring
contextuality for human measurements
cool
a second is understanding how context
switching
is implemented at the neurocognitive
level so that's
the feigning quote that we just looked
into but basically what is happening
when it's like somebody's your friend
you're in a narrative of friendship
and i get from the cultural down to the
you know the day down to the second down
to the
micro expression you're in a narrative
and then a stimuli happens and a context
switches
and that trajectory is still a unitary
your relationship with them is still a
thing
but it's changed in a way that the
context has changed like how does that
get implemented
this combination of the high precision

priors and the high precision inputs
how do we get the best of both worlds
with this rapid
attentional modulation as brains
into and away from objects and then
context switching
those are features that are just not
within
the scope of shannon disinfo theory and
they're much more semantic
questions a third and broader question
is whether intrinsic contextuality can
be detected and characterized at
multiple scales in complex systems
generally
by formulating a scale-free model of
contextuality in category theory terms
and in particular by employing the
concepts of classifiers and information
channels
we hope to have opened a new pathway
towards addressing these questions in
multiple ways
or addressing these questions in
multiple systems
so maybe if there is something special
to be said about contextuality
or the violations thereof two sides of
the same coin
maybe we could identify in a less
biased way what scales of organization
are relevant or just detect
new dynamics of complex systems
because there's a lot of recent work
showing that like with complex systems
they can emit
structured outputs that have the
signature of white noise

or have a signature of nothing like an
encrypted file
is going to have a distribution
syntactically
that's going to be looking like nothing
different from the frequentists
expectations but it's a semantically
encoded
local logic that is allowing
communication through that channel
so if all you have is the white noise
detector you're going to think up that
looks like the signal is not
containing information but actually
semantically it is
so there's there could be systems just
in and around us that are doing things
we don't even see
and that would be no weirder than if
there was wavelengths we didn't see we
know that's true
we know that there's things that are too
big or too small for us to see so could
there be things informationally or from
a narrative perspective
that we're just not perceiving
doesn't seem out of the picture so i
think it's a it actually is
a major area and you know for those who
have participated or listened to this
deep in
chris fields was the one who recommended
this paper to us and just said
i think this is a topic that is not
really thought about in the active
inference community that much but it
will become
important later i don't even know what

topic he was referring to but this was
the paper that
really raised some huge questions
so all kinds of questions that we have
but
i think it's fair enough just to say um
thanks to r2 3
brave and awesome participants thanks
sarah and blue because that was a fun
convo and we definitely laughed and
memed because
it was like the hardest paper i think
we've read yet
definitely but you know the paper was so
well written like i am not a big fan of
reading like math papers and
lemma this and theorem that like i'm not
i usually don't enjoy it at all but i
really did find
this paper to be very well written and
easy to follow
i agree when there was a formalism it's
a theorem
definition figure formalism is like okay
i'm not going to understand that but
i know that that's just the theorem so i
could just say okay
got it something i don't understand put
it away encapsulate that so i agree
um any last thoughts sarah blue
i just thought i was almost wishing like
as i was reading the paper like for some
concrete examples but
we would have ended up with like a 65
page paper
so i guess that's our job to be the
concrete example you know
yep and that will be fun to talk to

chris about so
um let's um until the
time we actually get to talk about this
paper which is going to be
on march 9th so between now and march
9th
let's just chill you know do our other
jobs
instead of just reading this paper think
about what would be really
powerful and meaningful to ask them
semantically